INCH-POUND

MIL-DTL-3928/30
w/AMENDMENT 1
24 July 2024
SUPERSEDING
MIL-DTL-3928/30
5 December 2014

DETAIL SPECIFICATION SHEET

SWITCHES, RADIO-FREQUENCY TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED) CLASS 5, 1P2T

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the switches described herein shall consist of this specification sheet and MIL-DTL-3928.

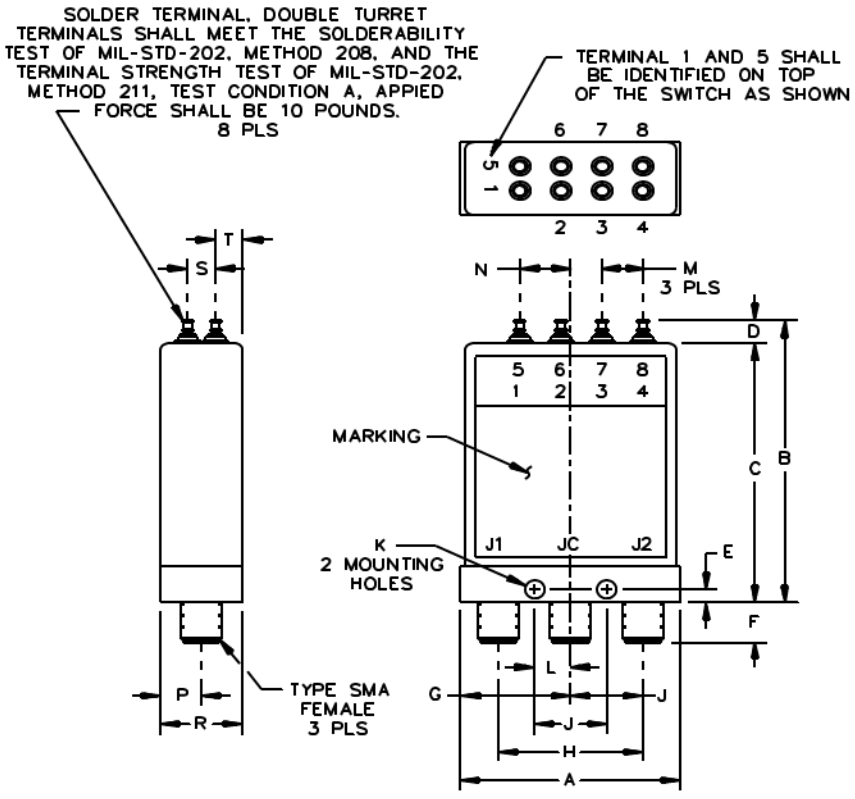


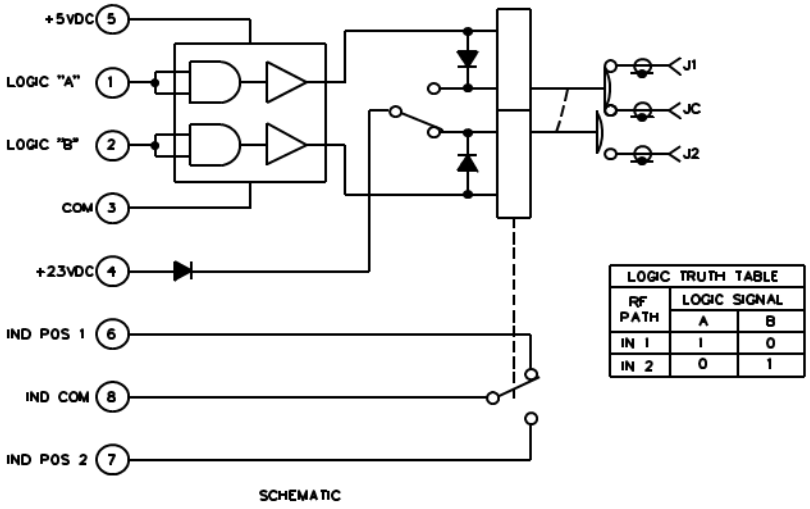
FIGURE 1. Switch configuration and schematic.

AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:21Z
Check the source to verify that this is the current version before use.



LOGIC TRUTH TABLE		
RF PATH	LOGIC SIGNAL	
	A	B
IN 1	1	0
IN 2	0	1

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
A 1.34 ± .030	1.31	1.37	33.3	34.8	K .120 ±.010	.110	.130	2.79	3.30
B 2.15 max	---	2.15	---	54.6	L .220 ±.010	.210	.230	5.33	5.84
C 1.81 ± .020	1.79	1.83	45.5	46.5	M .25 ±.030	.22	.28	5.6	7.1
D .35 max	---	.35	--	8.9	N.308 ± .032	.276	.340	7.01	8.65
E .09 ± .030	.06	.12	1.5	3.0	P .25 ± .030	.22	.28	5.6	7.1
F .28 ± .030	.25	.31	6.4	7.9	R .50 ± .032	.468	.532	11.88	13.51
G .670 ±.010	.660	.680	16.76	17.27	S .17 ± .030	.14	.20	3.6	5.1
H .88 ±.030	.85	.91	21.6	23.1	T .16 ± .030	.13	.19	3.3	4.8
J .440 ± .010	.430	.450	10.92	11.43					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents may be determined using 1.00 inch = 25.4 mm.

FIGURE 1. Switch configuration and schematic, part or identifying numbers M3928/30-01. Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:21Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and performance characteristics. 1/ 2/

Part or identifying number M3928/30-	Housing	Frequency range GHz	Frequency rating GHz	VSWR Max	Insertion loss (dB) Max	Isolation (dB) Min	Switch time (ms) Max	Position indication circuit and rating	Life cycles x 1000 Min 3/	Fail-safe or latching	Operating current (A) Max	Holding current (A) Max	Nominal operating voltage	Pickup voltage (less than)	Dropout voltage (less than)	Power and indicator connector	Weight (oz) Max
M3928/30-01 N,S	I	DC -18	DC -4.0	1.2:1	0.1	80	20	.5 A at 30 Vdc	1000	L	.16	---	23 ± 5% Vdc	20Vdc	----	8 solder terminals	5
			4.0-12.4	1.3:1	0.2	70											
			12.4-18.0	1.5:1	0.3	60											

1/ Average RF power: 15 Watts 0 to 10 GHz; 12 Watts 10 to 15 GHz; 10 Watts 15 to 18 GHz.
2/ Voltage rating: 350 Volts RMS.
3/ RF contact power switching 100 mW max.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:21Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.
Termination: Open.
Nominal impedance: 50 ohms.

RF connections: Connector shall be SMA Female configuration and shall meet the requirement of MIL-PRF-39012. The center contact shall be captivated and shall be capable of withstanding an axial force of 10 pounds minimum.

Electrical and performance characteristics: See table I.

RF contacts: Break before make.

Indicator contacts: Make before break.

Operating temperature: - 55°C to +95°C.
Storage: - 65ºC to +125ºC.

Drive Circuits:
Driver Voltage: 4.5 to 5.5 Vdc.
Driver current 70 MA max.

Logic “0” Voltage: 0 to .4 Vdc
Logic “1” Voltage: 2.4 To 5.5 Vdc with pulse width 20 ms Minimum.

Terminal: Terminals shall meet the solderability test of MIL-STD-202-208 and Terminal strength test of MIL-STD-202-211 test condition A, Applied force shall be 10 pounds.

Referenced documents: In addition to MIL-DTL-3928, this document references the following
MIL-PRF-39012
MIL-STD-202-208
MIL-STD-202-211

Amendment notations. The margins of this specification sheet are marked with vertical line to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - MI
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5985-2024-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:21Z
Check the source to verify that this is the current version before use.